

The commuting-pair joint-similarity question

A negative answer predating arXiv:1007.4655

Literature-status identification, lane 12 · 13 August 2026

Status. `literature_already_answered`. One of the two questions flagged in Loring–Shulman was already answered negatively by Pisier in 1998. The separate Olsen compact-perturbation questions are not claimed to be resolved here.

Source question

Loring and Shulman, *A Generalized Spectral Radius Formula and Olsen’s Question*, arXiv:1007.4655v2, state in their Introduction (printed p. 2) the question whether commuting Hilbert-space operators A, B , each individually similar to a contraction, must be simultaneously similar to contractions [1]. Their Theorem 8 proves the result when one member is similar to a strict contraction.

Decisive literature answer

Pisier’s paper *Joint similarity problems and the generation of operator algebras with bounded length*, published in 1998, states in its abstract that there is a commuting pair (T_1, T_2) such that each T_i is similar to a contraction while the pair is not similar to a pair of contractions [2].

Clouatre and Mbacke explicitly identify the same result in the abstract and Introduction of arXiv:1806.09027 (printed pp. 1–2) [3]. They cite it as Pisier’s Theorem 1 and record the stronger obstruction that $T_1 T_2$ is not even polynomially bounded.

Indeed, if one invertible Y made both $Y T_1 Y^{-1}$ and $Y T_2 Y^{-1}$ contractions, then

$$Y T_1 T_2 Y^{-1} = (Y T_1 Y^{-1})(Y T_2 Y^{-1})$$

would be a contraction. Hence $T_1 T_2$ would be similar to a contraction and therefore polynomially bounded, contradicting Pisier’s example. Thus the answer to the source question is *no*.

Chronology, provenance, and scope

The answer predates arXiv:1007.4655 by twelve years. This is therefore a literature-status correction, not a new counterexample. Clouatre–Mbacke’s paper knows and explicitly states the joint-similarity obstruction, cites the primary 1998 theorem, and contrasts it with their positive finite-dimensional result.

The original paper also asks Olsen’s two exact compact-perturbation questions: for a fixed polynomial, and simultaneously for all polynomials. Neither is answered by Pisier’s construction, so they are outside this packet. Exact phrase searches for Olsen’s question and inspection of the adjacent lifting papers arXiv:0907.2618 and arXiv:1111.6124 found no general resolution in the bounded search.

References

- [1] T. Loring and T. Shulman, *A generalized spectral radius formula and Olsen’s question*, J. Funct. Anal. 262 (2012), 719–731; arXiv:1007.4655v2.
- [2] G. Pisier, *Joint similarity problems and the generation of operator algebras with bounded length*, Integral Equations Operator Theory 31 (1998), 353–370, doi:10.1007/BF01195125.
- [3] R. Clouatre and D. Mbacke, *Joint similarity for commuting families of power bounded matrices*, arXiv:1806.09027v1, especially abstract and pp. 1–2.